

Opinion

Moving intentions from brains to machines

Christian Beste^{1,2,*}, Heleen A. Slagter^{3,4}, Christian Herff⁵, Yukiyasu Kamitani^{6,7}, Sabrina Coninx⁸, Richard van Wezel^{9,10}, and Christian Frings^{11,12}

Brain–computer interface (BCI) research has achieved remarkable technical progress but remains limited in scope, typically relying on motor and visual cortex signals in limited patient populations. We propose a paradigm shift in BCI design rooted in ideomotor theory, which conceptualizes voluntary action as driven by internally represented sensory outcomes. This underused framework offers a principled basis for next-generation BCIs that align closely with the brain’s natural intentional and action-planning architecture. We suggest a more intuitive, generalizable, and scalable path by reorienting BCIs around the ‘what for’ of action—user goals and anticipated effects. This shift is timely and feasible, enabled by advances in neural recording and artificial intelligence–based decoding of sensory representations. It may help resolve challenges of usability and generalizability in BCI design.

The necessity to shift concepts in brain–computer interface research

Brain–computer interfaces (**BCIs**; see [Glossary](#)) [1] hold transformative potential for restoring communication and action in paralysis in real time [2,3]. High-profile examples of assistive communication technologies exemplify how BCIs can re-enable autonomy and interaction, turning severe disability into **agency**. Yet one can ask if ‘agency’ is restored using current BCIs.

Consider, as an example, a paralyzed patient using a BCI operating a wheelchair or other forms of prosthetics [4,5] to move through a door from the kitchen into the garden. The patient’s goal is clear (‘I want to get into the garden’), but the way this is accomplished nowadays, using BCIs, is motor-oriented (‘exert a motor command to move the prosthetics’). Across psychology, neuroscience, and philosophy, agency is defined as the awareness of oneself as initiating and performing a voluntary movement, involving intentionality and decision-making. So, agency implies not just motor commands but motor commands with purpose (i.e., a goal) [6,7].

It is this intentionality that is currently lacking in the design of BCIs. Even when BCIs enable movement, users rate **embodiment** relatively low [8,9]. This suggests that current BCI systems restore motion, but not the sense of control central to agency. Most current BCIs are motor cortex–based (homunculus-driven) and biomechanics-oriented, with neural signals measured directly from the motor cortex using penetrating microarrays [8–10] or electrodes on the brain’s surface [10–12]. Alternatively, evoked potentials from observed stimuli corresponding to selected actions are employed [13]. Despite significant successes of both approaches, current BCIs are, by design, limited to decoding individual action primitives (e.g., ‘move left’ or a sequence of phonemes in a speech BCI), rather than goal-directed action sequences, and are unlikely to scale far beyond current capabilities. Furthermore, their clinical application remains restricted to patients with motor pathway disruptions in which information relayed from the motor cortex to the actuators is blocked, such as those with amyotrophic lateral sclerosis, brain-stem stroke, and very high spinal cord injuries. While actions are implemented through the motor cortex, many more processes

Highlights

Brain–computer interface development has advanced rapidly but remains largely anchored in decoding motor output or visual attention signals.

Current approaches often overlook how the brain naturally organizes voluntary actions around goals and expected sensory outcomes.

Ideomotor theory offers a powerful yet underused framework for aligning brain–computer interface design with action planning in the human brain.

Reframing brain–computer interface decoding to prioritize users’ intended effects rather than overt movements may enable more intuitive and flexible control.

Advances in neural recording and artificial intelligence decoding now make it feasible to realize ideomotor-informed brain–computer interfaces with broader usability and scalability.

¹Cognitive Neurophysiology, Department of Child and Adolescent Psychiatry, Faculty of Medicine, TU Dresden, Dresden, Germany

²German Center for Child and Adolescent Health (DZKJ), Partner Site Leipzig/Dresden, Dresden, Germany

³Department of Experimental and Applied Psychology, Vrije Universiteit Amsterdam, Amsterdam, The Netherlands

⁴Institute for Brain and Behavior Amsterdam, Vrije Universiteit Amsterdam, Amsterdam, The Netherlands

⁵Department of Neurosurgery, Maastricht University, Maastricht, The Netherlands

⁶Graduate School of Informatics, Kyoto University, Kyoto, Japan

⁷Computational Neuroscience Laboratories, Advanced Telecommunications Research Institute International, Kyoto, Japan

precede this, as we argue in the following based on ideomotor theory. Homunculus-driven BCIs at the end of this hierarchy possibly break down because motor-cortical readouts lack the fidelity to represent context-dependent goals, connections from intention-forming areas to motor regions are disrupted, or motor regions are damaged themselves. Direct decoding of intentions may restore goal-directed actions by bypassing impaired pathways. Notably, sensory and associative cortices often remain functional even when corticospinal pathways are disrupted [14]. Evidence from paralysis and amputation shows preserved, decodable sensory and body-related representations despite loss of motor output [15], and imagery-related activity in upstream planning networks remains accessible for decoding [5,16].

We outline how a paradigm shift can unlock further progress. Specifically, we propose that rethinking BCI design through the lens of ideomotor theory—emphasizing intention and goal representation (Box 1)—offers a powerful framework to drive this next development phase. The basic principles of this theory are shown in Figure 1.

According to ideomotor theory (Box 1), actions are represented in terms of what they achieve—that is, the motor features and ‘sensory effects’ that they generate. Human action planning thus relies on reafferent information: internally generated predictions of the sensory consequences of movement. Simply ‘thinking’ of (i.e., anticipating or activating) a wanted effect retrieves the action representation needed to bring it about. Movements can be controlled by anticipating the visual (or distal) effects they produce, even without feeling proprioceptive or somatosensory input. Current BCI approaches often neglect this fundamental architecture, focusing instead on low-level motor output that is cognitively demanding, unintuitive, and difficult to generalize. Ideomotor theory offers a principled alternative by targeting the level at which actions are naturally planned and selected in the brain: through their intended sensory effects. Grounding BCI design in these ideomotor principles can shift the field from a mechanistic focus on action output to a goal-oriented framework rooted in intention and outcome representation. We outline a conceptual roadmap for developing ideomotor-inspired BCIs, highlighting both the promise of this approach and the key challenges that must be addressed to realize it in practice. This roadmap is not a complete blueprint, but it offers a clear path forward that can guide future development through continued theoretical, technical, and empirical advances.

Box 1. Ideomotor theory and event files

Ideomotor theory is a long-standing framework in cognitive psychology [17] that explains how the anticipated sensory consequences of actions are used to retrieve action plans. It proposes that actions are guided by the internal activation of their anticipated sensory outcomes, offering a unified framework for understanding voluntary action, motor learning, and agency. Originating in the 19th century with thinkers like William James and further developed in modern cognitive science [17,18], the core idea is that actions are triggered not by abstract motor commands alone but by internal representations of their expected outcomes—what an action will look, feel, or sound like. In other words, action representations rely on reafferent information. Thus, when a person intends to act, they mentally activate or anticipate the sensory effects associated with that action (e.g., the feeling of the keypress and the visual or auditory feedback) [17–19]. This anticipation serves as a retrieval cue to initiate the corresponding motor activity, making action selection governed by outcome anticipation, not the motor plan (of which agents are typically unaware). This positions action control as goal-directed, tightly linking perception and action in a bidirectional loop. Ideomotor theory has broad implications for understanding actions. It explains automatic imitation and goal-directed behavior and suggests a bidirectional link between actions and their sensory outcomes. Later, concepts such as the Theory of Event Coding [18,20] and the Binding and Retrieval in Action Control concept [21–23] have been developed and extend ideomotor theory by providing a representational account of how these perception–action links are structured. An event file is a temporary, episodic binding of stimulus features, response features, and potentially contextual or goal-related information into a unified representation. When a part of this file is reactivated (e.g., seeing a stimulus), the entire file can be retrieved, including the associated action. Event files operationalize ideomotor theory and describe how ideomotor principles are implemented cognitively. Effect-based action control in ideomotor theory relies on the same feature bindings that event files describe. If an anticipated effect is re-perceived, the event file linking that effect to a specific action can be reactivated, leading to faster or more automatic responses.

^aDepartment of Philosophy, Vrije Universiteit Amsterdam, Amsterdam, The Netherlands

⁹Donders Institute for Brain, Cognition and Behaviour, Radboud University, Nijmegen, The Netherlands

¹⁰OnePlanet Research Center, Radboud University, Nijmegen, The Netherlands

¹¹Institute for Cognitive & Affective Neuroscience (ICAN), University of Trier, Trier, Germany

¹²Department of Psychology, Cognitive Psychology, University of Trier, Trier, Germany

*Correspondence:
Christian.beste@ukdd.de (C. Beste).

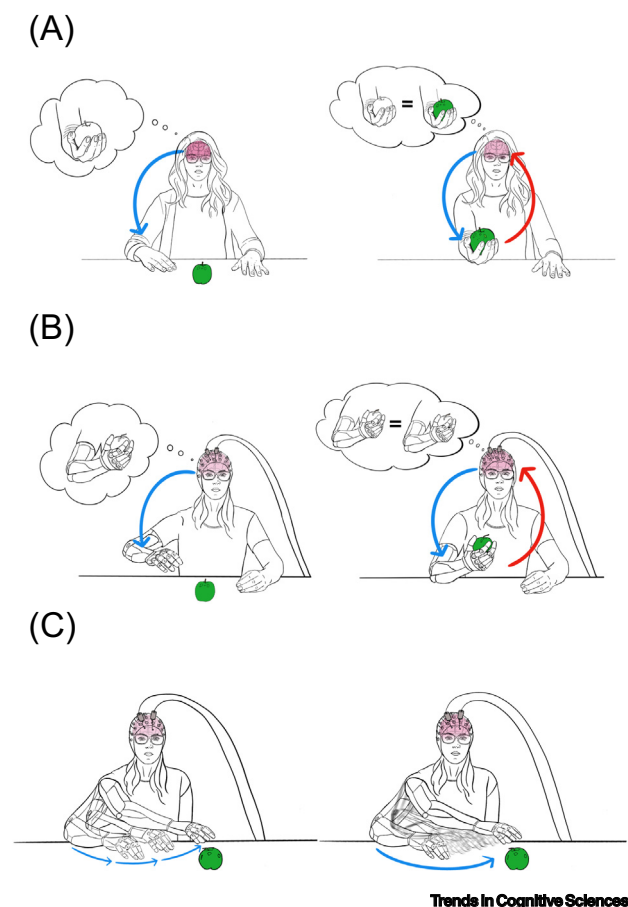


Figure 1. Ideomotor principles applied to BCI. (A) The principle of effect anticipation by the ideomotor theory. When one wants to grasp an apple, one anticipates its look and feel, leading to the retrieval of the motor program for grasping and executing it until the sensory effect (the perceivable effect in the environment) matches the anticipated effect. (B) The same principle might be used for motor control of a robot arm using BCI. (C) Goal-directed behavior leads to smooth movements (left part). In contrast, behavior with a biomechanical focus, as currently used in BCIs, leads to choppy motion (right part). BCI, brain-computer interface.

Glossary

Active inference: a theory of action, perception, and learning, where agents actively shape their environment to reduce uncertainty and maintain a coherent internal model of the world.

Agency: the capacity of an individual to initiate, control, and direct actions intentionally toward goals.

Artificial intelligence (AI) model: a computational system trained to perform tasks that typically require human intelligence (e.g., recognizing patterns, making decisions, or generating language). It learns from data and encodes relationships or rules to make predictions, classifications, or create outputs.

BCI: a system that enables direct communication between the brain and an external device, bypassing traditional neuromuscular pathways.

Embodiment: this concept emphasizes that the mind is connected to the body and shaped by sensorimotor experiences, bodily states, and the physical environment.

Forward model: an internal representation that predicts the sensory consequences of motor commands (e.g., predicting limb position when reaching) without waiting for feedback.

Goal-conditioned reinforcement learning: reinforcement learning is where an agent learns policies explicitly conditioned on a goal. Instead of learning to maximize reward in a fixed task, the agent is trained to generalize across multiple tasks and adapt to new goals without retraining.

Hierarchical autoencoder: a neural network that compresses and reconstructs data using multiple layers of abstraction, where each level captures increasingly abstract features. It learns complex patterns in a structured and interpretable way.

Local field potential: an electrophysiological signal reflecting the summed synaptic and dendritic activity of local neuronal populations.

Multisite activity: the activity of multiple single neurons is typically measured with penetrating electrodes.

Oscillatory activity: rhythmic electrical patterns are generated by populations of neurons characterized by their frequency, amplitude, and phase and are thought to play key roles in coordinating brain functions.

Reinforcement learning: an agent takes actions, observes the resulting

From motor commands to action goals: a framework for intention-based BCIs

A key insight from ideomotor theory is that action is fundamentally goal-oriented, not merely about motor execution. In natural behavior, no two movements are ever the same, but what remains stable is the intended outcome. Even highly trained athletes never repeat a movement in precisely the same way. Their skilled behavior is grounded in achieving consistent outcomes despite variability in muscle movement. Such variability allows for flexibility at the motor level in the light of a dynamically changing environment and stability at the higher functional level. Pedagogical models, such as the Constraints-Led Approach, leverage this principle by creating learning environments that encourage various movement solutions to achieve the same invariant goal [24]. This underscores a central design principle for ideomotor-based BCIs: systems should prioritize decoding the user's intended sensory outcomes, which are expected to be stable across varying movements, rather than specific motor commands. Although sensory imagery may vary phenomenologically, neural representations in sensory cortices are highly stable across contexts [25,26]. While percept-evoked stability is strongest, imagery elicits highly similar, decodable patterns in early and high-level sensory cortices [27–29], likely enabling reliable decoding of anticipated sensory outcomes and supporting its suitability for ideomotor BCIs. This makes effect-based decoding less susceptible to drift or fatigue, possibly supporting greater robustness and generalization.

Intentions also structure action hierarchically. The same motor act, such as 'typing', can serve several higher-order goals, including 'writing a report', 'working', 'trying to meet a deadline', or

'helping a team finish a project'. These layers of intention are intertwined: higher-level characterizations explain why an action is performed (e.g., trying to meet a deadline), while lower-level characterizations describe how it is carried out (e.g., typing). This structure enables action stability across variable motor implementations. Through higher-level characterizations, one may feel more strongly that one is acting on one's goals, rather than just reacting.

This principle is especially relevant for BCI-controlled prostheses [5], where the sense of embodiment—the feeling that the prosthesis is 'part of me'—depends less on biomechanical fidelity and more on whether the system supports goal-oriented, intuitive control [30]. BCI-controlled prostheses do not need to imitate the body's exact motoric processes. However, they must enable equally good ways to interact with the world, allowing the person to reach the same goal [31]. A BCI controlled at a higher level of functionality that focuses on the purpose of an action may increase the feeling of embodiment, a combination of **sense of ownership** and **sense of agency**. Supporting this, neuronal responses in the human primary motor cortex have been shown to coincide with the subjective onset of movement intention in BCI-mediated actions [32].

The ideomotor theory furthermore holds that simply imagining the sensory outcome of an action is sufficient to trigger an action plan. For BCIs, this implies that the system should decode the intention or the anticipated action outcome, not just raw motor signals, using neurophysiological data. Momentum is building for intention-based decoding approaches in cognitive neuroscience [33–35], a necessary step for ideomotor BCIs. While earlier high-level or shared-control BCIs [36,37] also bypass low-level motor commands, they remain data-driven. In contrast, the ideomotor framework specifies the representational level to be decoded, which adapts to the user. Recent studies along ideomotor theory show that directed communication in temporo-frontal networks underlies action–effect anticipation [38–41]. These networks activate before motor execution and remain stable during action execution [38], making them promising targets for intention decoding, together with sensory and parietal cortical areas [16,42]. This is consistent with the ideomotor principle, which incorporates a **forward model** [43] and context-dependent learning in a feedback loop [44,45], whereby anticipated sensory consequences of an action are used in action chains. From a learning perspective, ideomotor control emerges through repeated associations between actions and their sensory effects. Early stages link proximal outcomes to movement, while later stages generalize to distal goals [46,47]. In this way, ideomotor theory naturally incorporates predictive processing at sensorimotor levels. As such, it aligns closely with predictive coding frameworks [48], including **active inference** [49], reinforcing the idea that to design BCIs consistent with how the brain plans and initiates action, we must prioritize decoding intentions and anticipated outcomes over motor execution.

Redefining neural targets: BCIs aligned with the brain's planning architecture

If we want BCIs to work in harmony with the brain's natural planning and control of action, we need to rethink the types of brain signals we use and how we train people to utilize them. In this section, we show how ideomotor principles shift the focus from decoding physical movements to decoding the user's intended outcomes, and what this means for selecting brain signals, designing training, and building more intuitive, goal-directed BCIs.

Typically, BCI technology is fitted to the user at the beginning of BCI usage by collecting brain activity during the targeted behavior (e.g., attempting an arm movement). Features of the recorded neural activity are then used to train a decoder. As the patient starts using the BCI, the brain adapts to the BCI technology, which might also update its initial decoder weights. This process is called coadaptation [50] and is crucial for high performance [51]. Currently, most BCI researchers use features of brain activity (e.g., **oscillatory activity**, **local field potential**, and

states, and receives rewards based on actions. The goal is to learn a policy that maximizes the cumulative reward over time. Learning occurs through trial and error.

Sense of agency: one's experience of being the author of a movement.

Sense of ownership: the experience of being the subject of the movement, voluntarily or involuntarily.

Single-unit activity: the activity of one single neuron is typically measured with penetrating electrodes.

World model: an internal representation that an agent learns to predict how the environment behaves. It simulates the consequences of actions by predicting future states and rewards based on current states and actions.

multiunit activity or **single-unit activity**) ‘that work’, and this choice is often uninformed by the ideomotor principles. Instead, practical considerations, such as ease of implantation [52] or markers of decoding quality, guide the decision. In practice, this means that a limited number of brain signals are used depending on the recording technique employed. If at all, current BCI technology focuses on the retrieval part of ideomotor principles, but the ‘binding part’ or learning of the motor pattern is largely spared, as mostly neural activity concurrent to motor primitives is used during BCI training. The way we train people to use technology must be improved.

As noted previously, technically implementing ideomotor principles in BCIs necessitates robust methods for decoding the intended sensory consequences of an action, rather than just the motor commands themselves. The shift toward ideomotor BCIs redefines the primary decoding target. Instead of focusing solely on neural signals from the motor cortex that correlate with specific movements (the ‘how’), the emphasis shifts to decoding neural representations of the intended sensory state or action effect (the ‘what for’). Advances in brain decoding, particularly techniques for reconstructing perceptual content from brain activity in sensory cortices, provide a strong foundation [27,28]. These methods demonstrate the feasibility of extracting anticipated visual, auditory, or somatosensory outcomes. The challenge lies in translating these decoded sensory goals into effective control signals for external devices. This requires accurate decoding and sophisticated **artificial intelligence (AI) models** that can bridge the gap between a predicted sensory state and the sequence of actions needed to achieve it. While our understanding of the precise neural mechanisms underlying the formation and retrieval of action–effect associations (event files) is still evolving, computational models inspired by active inference and **reinforcement learning** are beginning to offer plausible frameworks to implement these insights into technical solutions (Box 2).

Box 2. Operationalizing ideomotor theory in BCIs: active inference, reinforcement learning, and world models

Ideomotor theory is conceptually aligned with predictive, inferential frameworks, particularly active inference. According to active inference, a computational framework grounded in the free energy principle, actions are generated to minimize the difference between predicted and actual sensory input [53]. Like the ideomotor theory, it highlights the role of anticipated sensory outcomes in guiding goal-directed behavior but uniquely casts this process as inference over future sensory states. We focus on ideomotor theory because of its strong empirical foundation, including at the level of neural implementation, and psychological clarity, especially in describing how intentions are formed, maintained, and linked to anticipated effects (Box 1). Yet ideomotor theory lacks a formal computational implementation. Active inference may offer a computational complement, providing the tools to infer and update user intentions in real time. Deploying active inference models in online BCIs remains technically demanding because model optimization and parameter adaptation must occur under strict temporal constraints. Earlier implementations [54] have demonstrated feasibility but used active inference as an adaptive controller. An ideomotor BIC can build on this by specifying the target of inference, thereby constraining model updating.

Reinforcement learning—while powerful for optimizing reward-based behavior—does not model action as driven by predicted sensory outcomes but frames action selection around reward maximization [55]. Nonetheless, reinforcement learning could complement the optimization of long-term performance, particularly in hierarchical or **goal-conditioned reinforcement learning**. Indeed, ideomotor theory naturally supports a hierarchical view of action goals. A high-level intention (e.g., make tea) can be decomposed into subgoals (e.g., boil water, get a cup, add a tea bag), each associated with anticipated sensory consequences. This perspective maps well onto hierarchical reinforcement learning, where policies are learned at multiple levels of abstraction [56]. Similarly, goal-conditioned reinforcement learning offers a powerful implementation strategy, particularly when combined with learned world models [57]. These models learn a representation of the environment’s dynamics and use this to ‘imagine’ the consequences of different action sequences, selecting those that lead to desired states (goals). Integrating principles from ideomotor theory, such as the bidirectional learning of action–effect associations and goal representations, could further enhance their capabilities, making them more human-like. For example, world models could be trained to learn the mapping between actions and their intended versus actual sensory effects, rather than to predict future sensory inputs, refining internal models through a process akin to event file formation. In ideomotor BCIs, world models act as adaptive generative agents that learn bidirectional mappings between decoded intentions and predicted sensory outcomes. Decoded goal states from cortical networks are used to simulate expected feedback and update control trajectories. Technically, recurrent state-space models or variational autoencoders can maintain a latent variable capturing anticipated sensory consequences, refined through Bayesian updating, in line with active inference principles.

In principle, any form of neural activity can be used as a ‘tool’ to fulfill ideomotor goals. However, two key considerations should guide the selection of neural targets: (i) technological feasibility and signal quality for decoding and (ii) how readily the activity can be learned and voluntarily modulated by the user. This raises the crucial question of what the decoding target should be. Rather than decoding low-level motor commands, ideomotor theory suggests BCIs should target neural representations of the anticipated sensory consequences of an intended action. These representations may reside in parietal regions involved in action planning or in sensory cortices encoding the anticipated visual, auditory, or tactile action effects [27,29,58,59]. For example, the intention to grasp an apple might be more effectively decoded by identifying neural activity that reflects the apple’s expected visual and tactile properties, rather than the specific motor program for grasping. Moreover, users often imagine multiple potential outcomes or sensory scenarios when forming an intention. To support intentional specificity, the BCI system must be capable of disambiguating among these competing representations and targeting the specific sensory outcome the user envisions. This precision is critical for eliciting a genuine sense of agency, allowing users to feel that BCI-mediated actions truly reflect their volition. Achieving this may require multiple recording locations to capture the distributed outcome representations. Recent advances in biomedical engineering reveal that various features in neural activity can reliably be extracted without adjustments in the analytical processes to decode movement intentions [60]. This supports the feasibility of targeting higher-level action goals. Amodal action representations are primarily integrated within premotor and parietal cortices [61–63], where amodal action–effect codes combine effector-independent and sensory features, complemented by sensory areas encoding anticipated outcomes. From an ideomotor perspective, decoding signals from regions of the aforementioned regions, as well as sensory cortical areas, parietal cortex [64,65], hippocampal structures [66], or other nodes in the action-mode network [67], may offer direct access to user intentions as represented through anticipated sensory consequences.

To align BCI design with ideomotor principles, it is crucial to move beyond the reliance on single-site and single brain signal–type strategies. Ideomotor-inspired approaches require a more nuanced view of the brain’s representational and planning architecture that reflects how intentions are dynamically encoded across distributed networks and multiple brain signals. Emerging evidence supports this shift [21]. Ideomotor theory specifies which distributed signals matter, providing a principled, goal-based rationale. Studies suggest that event file content is captured by beta-band activity [68,69], while processes related to binding and retrieval are primarily associated with theta/gamma-band activity [70–72]. These dynamics are likely controlled by alpha-band activity [73]. This is important because alpha band activity likely serves predictive coding processes during event file management [74] and is therefore essential for planning. This implies that ideomotor-aligned BCIs must integrate activity across multiple frequency bands and multi- and single-unit activity to capture the full scope of intention-related neural processes. This may provide a richer, temporally structured readout of the brain’s internal action models.

Finally, event file formation and adaptation are network-level processes [38,73] with a distinct temporal profile [75,76]. The parietal cortex [77–79], together with areas in the ventral stream pathway [80] and the premotor cortex [81], is involved during event file coding. At the same time, temporo-frontal networks are engaged in action–effect anticipation and learning [38]. Successful implementation of ideomotor BCIs will thus require decoding distributed network activity, in contrast to current BCIs, which typically focus on motor cortex activity during motor execution. While the motor cortex remains a highly effective site for signal extraction, relying solely on a single region may miss the broader, goal-oriented representations [82] central to ideomotor theory. Whether it is better to focus on specific aspects of oscillatory activity, multiunit activity, specific

neuroanatomical regions, or a combination of them remains an open question that must be addressed together (see [Outstanding questions](#)).

Modern BCIs already identify informative electrodes and frequency bands across cortical sites. The ideomotor framework complements this by predicting that the most informative features are those reflecting anticipated sensory effects and integrated action–effect codes. This narrows the search space, enabling more interpretable, generalizable, and potentially powerful BCIs, as the narrowed search space confers algorithmic advantages during data processing. Ideomotor principles apply to both invasive and noninvasive BCIs, but with different emphases: invasive systems access high-resolution motor representations, which are suited for hierarchical goal translation, whereas noninvasive systems capture distributed activity, which is better suited for intention decoding. Incorporating neural ideomotor principles into BCI designs may have fundamental consequences, as discussed in section Generalization, abstraction, and compositionality in ideomotor BCIs.

Generalization, abstraction, and compositionality in ideomotor BCIs

As discussed in [Box 2](#), ideomotor theory provides a strong representational target, while active inference and reinforcement learning offer tools to implement intention-based BCIs. Yet a major challenge lies in developing BCIs that can generalize learned action–effect associations beyond the training context. This problem closely mirrors the compositionality problem in AI: how can an agent learn basic building blocks (primitives) and combine them systematically to solve new tasks [\[83,84\]](#)? For ideomotor BCIs, this means learning ‘abstract’ or ‘contentless’ event files that capture the general structure of an action–effect relationship, which can then be instantiated with specific parameters for a given context. For example, an abstract ‘reaching’ event file could be combined with a decoded target location (a specific sensory goal) to generate a context-appropriate reach. Achieving this depends on AI architectures that learn and manipulate hierarchical and compositional representations through modular neural networks or symbolic reasoning components integrated with deep learning. **Hierarchical autoencoders** and other deep learning architectures that learn sparse, disentangled representations [\[85\]](#) are promising. These models learn to transform rich, high-dimensional sensory inputs and motor outputs into low-dimensional, abstract latent codes that capture the essence of an action–effect contingency. For BCIs, this could mean learning a hierarchy where lower levels represent fine-grained sensory details or motor primitives, while higher levels represent more abstract goals or action schemata. Decoding at higher, more abstract representational levels could improve robustness and generalization, making BCI control less sensitive to environmental variation or execution-level noise.

World model AIs [\[86\]](#) are especially promising as they learn predictive models of how actions influence future sensory states. In such systems, a BCI might first decode a user’s intended sensory goal, leveraging techniques such as modular decoding [\[27,87\]](#). This decoded goal could then serve as an input to the world model to plan a sequence of actions to achieve that sensory state by ‘imagining’ trajectories within its learned model. This two-step approach directly addresses the need for BCIs that capture the agent’s intention at the desired outcome level.

A significant challenge for BCI generalizability across sessions for a single user and different users is the variability of neural codes. Neural code conversion aims to address this by learning transformations that map neural activity patterns from one domain (e.g., subject A or subject B at time 1) to another (e.g., subject B or subject B at time 2), ideally preserving the encoded intentional or sensory information [\[88,89\]](#). In the context of ideomotor BCIs, this could mean developing models that can align representations of intended sensory outcomes across individuals, even if their underlying neural encodings differ. Such an approach could drastically reduce calibration times and pave the way for more universally applicable BCI systems, potentially by leveraging

large-scale datasets to learn common underlying structures in how brains represent goals and anticipated effects.

While few current AI architectures are explicitly labeled as ‘ideomotor networks’, the core tenets of ideomotor theory resonate strongly with several leading AI paradigms. Specifically, world model-based reinforcement learning [90] embodies the ideomotor principle of action selection based on anticipated outcomes. These architectures offer a promising computational foundation for next-generation BCIs that decode users’ intended sensory goals and plan actions accordingly (Box 2), although technical challenges remain to be addressed (Box 3).

Ethical dimensions of ideomotor-based BCIs

Ideomotor-based BCIs raise unique ethical challenges beyond those typically associated with motor-based systems. First, these systems aim to decode internally generated goals. Yet not all mental imagery reflects a true intention to act. This raises two interrelated concerns: the risk of mistakenly acting on fleeting or ambiguous thoughts and the broader issue of mental privacy. Without clear user consent or confirmation, these systems may expose or act upon intentions users never meant to express. To address this concern, BCI systems must incorporate robust safeguards that give users control over when and how intentions are acted upon. This could include requiring a second confirming signal (e.g., eye movement or distinct neural pattern), brief hold periods to verify intention persistence, or the ability to issue a cancel command.

Another practical safeguard is an adaptive thresholding mechanism that requires intention-related activity to reach temporal and spatial stability before output, preventing transient

Box 3. Technical challenges and possible solutions

Operationally, an ideomotor BCI decodes neural patterns representing anticipated sensory consequences. These decoded states feed into a generative model that computes the most efficient trajectory. Realizing such ideomotor-inspired BCIs requires addressing challenges. These include moving beyond traditional biomimetic training toward goal-directed learning, decoding distributed multiplexed signals, and integrating real-time spatio-temporal neural data with generative AI models capable of translating abstract goals into actionable control signals. Current BCI training often relies on instructed tasks (e.g., ‘imagine moving your hand’), limiting self-initiated goal setting. Shifting to volitional, goal-based learning is essential for capturing genuine intention and enabling more natural BCI control that generalizes better [91]. A further challenge arises from the distributed and multiplexed information processing in frequency bands and brain regions [21,92]. According to ideomotor principles, BCI design must adopt a spatio-temporal neuroscience approach [93] and combine this with real-time data analysis [3,94]. Computationally, ideomotor BCIs can be implemented using active inference frameworks [53] and reinforcement learning to learn and refine action–effect contingencies. Once an abstract goal or intended sensory state is decoded, generative AI models (e.g., Generative Adversarial Networks or Diffusion Models [95]) may translate this abstract plan into a sequence of control signals for a prosthetic [96]. Ideomotor control depends on reciprocal information flow between intention and sensory consequences. In current BCIs, this loop is limited by sparse or delayed feedback. Although reproducing detailed proprioceptive signals remains challenging, hierarchical bidirectionality can be approximated by coupling fast, predictive updates from the machine (e.g., estimated limb state or goal proximity) with low-latency multimodal feedback.

Practical constraints to be navigated refer to the trade-off between signal richness and reliability, as well as the limitations of current recording technologies [ranging from focal invasive implants with penetrating electrodes to electrocorticograms (ECoG) vs. distributed scalp-based recordings]. Current recording technologies are often misaligned with the ideomotor theory’s emphasis on distributed, network-level activity. High-performance BCIs typically use invasive implants targeting motor regions and yielding clean signals, but they are not used for capturing goal-related representations distributed across brain areas. In contrast, noninvasive methods offer broader spatial coverage and access to cross-regional dynamics, consistent with ideomotor principles. However, they suffer from noise. Bridging this gap may require next-generation electrodes. Recent advances in high-density electrodes, micro-ECoG recording systems [15], and new nanotechnologies (e.g., nanowire-based devices) combined with miniaturized optical and ultrasound on-chip technologies may lead to solutions. For users with long-term paralysis, the ability to simulate sensory outcomes may be weakened, but see other recent work [14,97]. Ideomotor BCIs can include preparatory calibrations pairing imagery or external sensory cues with neural recordings to relearn action–effect associations. As consistent mappings re-emerge, the decoder adapts, and control gradually shifts to internal prediction. Such retraining may enable control even in severely impaired patients.

activations while preserving fluent control. At the societal level, although current BCI technologies do not yet enable ‘mind-reading’, the prospect of decoding internal states raises concerns about potential misuse, for example, by authoritarian regimes seeking to infer or manipulate mental states. As the technology advances, safeguarding against such misuse will become increasingly important, requiring consent protocols, secure data handling, and international ethical governance frameworks.

Another ethical challenge concerns accountability. If a system misinterprets an intention and acts on it, questions arise about who bears responsibility: the user, the system, or its designers. This complicates notions of agency and authorship and requires that BCIs are designed with built-in transparency and accountability. For example, the system could log which neural signals led to which actions, make its decision-making process transparent and interpretable to users and developers, and allow users to override or reverse system actions if needed.

Furthermore, adaptive BCIs that shape user behavior through feedback may unintentionally constrain autonomy, subtly nudging users toward specific outcomes. Addressing this requires continuous monitoring of adaptation processes and possibly the incorporation of user-defined boundaries for how the system may learn or respond. Finally, how techniques that improve function impact people’s overall quality of life is a complex matter that cannot be reduced to an improvement in action. For example, in Parkinson’s disease, we see that an improvement in motor function after implantation of a deep brain stimulation device does not necessarily come with an improvement in quality of life, as patients struggle with reintegration into family and social life. Ethical evaluation of ideomotor BCIs should include broader assessments of user well-being and lived experience.

Concluding remarks

We propose an ideomotor shift in BCI design that prioritizes the decoding of anticipated sensory effects of actions, rather than low-level motor commands, as current BCI systems typically do. This perspective reflects a fundamental principle of human action: that movements are organized around their goals, not their biomechanics. Yet most current BCIs overlook this goal-directed structure, limiting their intuitiveness, adaptability, and generalizability across users and contexts. This ideomotor perspective opens up new research directions, including the development of computational tools for intention decoding, the redesign of training protocols and hardware for goal-based control, and the exploration of the implications for agency, ownership, and user experience. Such research may improve BCI performance and also provide a deeper understanding of how goals and intentions are represented and enacted in the human brain in more naturalistic contexts. We hope that the roadmap outlined in this article will inspire interdisciplinary efforts to realize this potential.

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Outstanding questions

How can ideomotor theory be formalized computationally to inform BCI design?

What are the neural mechanisms of action–effect binding and retrieval in naturalistic contexts?

Which neural features best reflect intentional, goal-directed states: frequency bands, multi- and single-unit signals, anatomical regions, or their interactions?

How stable and transferable are intention-related codes across time, tasks, and users?

Can we develop world model-based BCIs that infer and plan actions from decoded sensory goals?

How can BCIs learn compositional and hierarchical action representations?

Can we design BCIs that support internally generated, self-initiated goals rather than externally cued tasks?

Which recording methods best capture distributed, multimodal representations of intentions?

Can generative AI models help map sensory goals to BCI control commands?

How can agency and volition be preserved or enhanced in ideomotor BCIs?

How can BCIs distinguish between imagined effects, intentions, and fantasies in neural signals?

How can BCIs dynamically disambiguate between multiple competing sensory goals in real time?

What ethical frameworks are needed to ensure user agency, privacy, and accountability in intention-decoding BCIs?

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